

# Abishek Soti

Sydney, Australia | [linkedin.com/in/abisheksoti](https://www.linkedin.com/in/abisheksoti) | [github.com/AbishekSoti](https://github.com/AbishekSoti) | [scholar.google.com/citations?user=HF4dEmSb\\_SQC](https://scholar.google.com/citations?user=HF4dEmSb_SQC)

## Education

### Western Sydney University

*Master of Philosophy, Machine Learning / Embedded AI*

- Research: underwater acoustic classification, event cameras, neuromorphic sensing, embedded AI, and model deployment.

Sydney, Australia

Aug. 2025 - Present

### Western Sydney University

*Bachelor of Engineering (Honours), Electrical Engineering; GPA: 3.902*

- Strong quantitative coursework: Electrical Fundamentals, Engineering Computing, Database Design, Digital Systems, Biomedical Signals/Data Analysis.

Sydney, Australia

Completed July 2025

## Technical Experience

### MPhil Researcher - Machine Learning and Signal Processing

*International Centre for Neuromorphic Systems, MARCS*

- Build Python pipelines for real-world acoustic datasets, covering preprocessing, feature extraction, model training, evaluation, and benchmarking.
- Develop ShipsEar audio classification workflows and compare model behaviour across classical ML and PyTorch deep-learning approaches.
- Use quantitative validation habits: define assumptions, compare methods, inspect failure cases, and communicate technical trade-offs.

Aug. 2025 - Present

Western Sydney University

### Research Assistant - Embedded and Event-Based Systems

*International Centre for Neuromorphic Systems, MARCS*

- Develop Python analysis workflows for event-camera data, including timestamped event processing, anomaly checks, and research reporting.
- Deployed ML inference on Raspberry Pi and monitored system resources, connecting algorithm behaviour with constrained runtime conditions.
- Troubleshoot systems across sensors, Arduino control, battery/power circuits, LED drivers, Raspberry Pi Linux, and SSH deployment.

Sep. 2024 - Present

Western Sydney University

## Projects

### FPGA / Digital Hardware Design

- Implemented digital hardware projects using VHDL/Verilog, timing/data-capture workflows, and Vivado-based development.

### ShipsEar Underwater Acoustic Vessel Classification

- Built an audio preprocessing, feature extraction, evaluation, and benchmarking workflow for vessel classification research.

### Production Engineering Learning Track

- Focusing on Docker, FastAPI, AWS EC2/ECR/S3, MLflow, ETL, CI/CD with GitHub Actions, deployment, and MLOps.

Current

## Leadership and Communication

### Customer Service Coordinator

*Unilodge*

- Support a 1,600+ resident community, owning issues through resolution, escalating risks, training staff, and coordinating with stakeholders.
- Prepared SOPs and routine reports using SQL/internal systems, strengthening documentation, communication, and operational follow-through.

Nov. 2021 - Present

Sydney, Australia

## Technical Skills

**Languages:** Python, SQL/SQLite, MATLAB, VHDL/Verilog, basic HTML/CSS/JavaScript, some C

**Software Fundamentals:** OOP, data structures and algorithms practice, logging, exceptions, regex, file handling, context managers, Git/GitHub

**ML / Quant:** PyTorch, scikit-learn, NumPy, pandas, matplotlib, Jupyter, model evaluation, signal processing, benchmarking

**Systems:** Raspberry Pi inference deployment, Linux/SSH, Arduino, sensors, event-camera pipelines, FPGA/digital systems